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LETTER TO THE EDITOR

Comment on: "No Time to Lift? Designing Time-Efficient Training Programs for Strength and Hypertrophy: A Narrative Review"

Luke Carlson[1] · Dave Smith[2] · James P. Fisher[3]

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Dear Editor,

We read with interest the recent review by Iversen and colleagues titled "*No Time to Lift? Designing Time-Efficient Training Programs for Strength and Hypertrophy: A Narrative Review*" [1]. We commend the authors on their efforts toward emphasizing the health benefits and importance of resistance training, and for providing these time-efficient recommendations. However, we would like to broaden the application of this article and identify further points worthy of discussion.

Specifically, the authors state, "*This narrative review is intended for those in the general public that have limited time for training, and not for those who are seeking to optimize training adaptations without regard to a time commitment (e.g. athletes)*." We appreciate that Iversen et al. [1] were not addressing an *optimal* approach to strength training for an athletic population. However, we should not assume that simply because athletes are looking to *optimize* adaptations that they have additional time to commit to strength training. Indeed, we should not underestimate the time constraints that exist for athletic populations. In our experience, athletes present the paradox of: (1) having much to gain from resistance training, i.e., injury prevention and athletic/performance adaptations [2] as well as the health benefits discussed by Iversen et al. [1], while (2) having very little time to engage in resistance training relative to the general population. Our observation may be equally true for professional athletes, collegiate athletes, high school athletes, and high level, adult, amateur athletes. The current authors present coaching and strength training experience with national and international (including Olympic and Paralympic) sports teams and athletes. According to that experience, fitting in strength training workouts with the demands of other conditioning sessions, competition, travel, meetings, film sessions, technical/skills practice, treatment/therapy, media events, and community events as well as other restraints, such as the union collective bargaining agreement, presents an ongoing challenge for athletes and strength and conditioning staff. This schedule only intensifies in collegiate athletics with the addition of academic course load and more frequent travel and competition, sometimes including two to three weekly competitions based on the sport. Further, a high school athlete deals with even more time spent in the classroom and has the added complexities of limited access to strength-training facilities compared to professional and collegiate athletes (the weight room may not be open, and a strength coach may not be available when the athlete has a respite from class, studying, practice, games, and part-time jobs). Finally, additional time constraints on strength training exist in the form of full-time employment, dependants, spouses, and other familial and social relationships, which are not defunct simply by being termed an *athlete*, even an elite one. These are not only important in considering Olympic and Paralympic athletes, but also the weekly regimens of amateur marathon runners or Ironman triathletes, whereby the training volume and time commitment leave little time for strength training.

Based on the above commitments and stresses, we should recognize that, in contrast to the general public, athletes suffer from burnout, stress fractures, and overtraining [3–5], and (with the exception of bodybuilders, powerlifters, and Olympic weightlifters, where strength training effectively is the sport) are primarily dependent upon the application of strength in performance of a fine motor skill. In this sense,

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athletic populations might also be suited to a time-efficient, minimum effective training dose to avoid physiological and psychological detriment, and to allocate more time to the application of their muscular strength in the skills required within their sporting endeavor. Indeed, recent data have supported that even well-trained athletes can make increases in muscular strength as a result of a time-efficient training program [6].

The authors also advocate the use of free-weight compared to machine-based resistance exercises, stating that "… *it is easier to simulate real-life movements and sport-specific movements with free weights compared to machines, which usually have limited adaptability of the movement pattern.*" Whilst we agree that resistance machines are limited in their movement pattern, typically only moving in one plane, we suggest this is more of an advantage. Research has suggested that muscular adaptations might be delayed and even less pronounced when using free weights compared to resistance machines [7]. This appears to be due to the need to learn the technique of balancing and coordinating free-weight exercises (i.e., the skill), compared to a more gross motor skill that requires force in only one fixed plane (i.e., a resistance machine). Further, it is noteworthy that if time efficiency is of primary importance, then the loading and unloading of a barbell can be avoided by using selectorized resistance machines that require only the movement of a pin in a weight stack and the adjustment of a seat position.

In closing, we do not disagree with most of the authors' recommendations [1]. However, we propose that, in view of the inordinate time commitments of athletes, the time-efficient, minimal effective dose-inspired guidelines that the authors provide for the general population can be extended to include the athlete population too, and the strength coaches who work with them, where appropriate.

Declarations

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Conflicts of Interest Luke Carlson, Dave Smith and James Fisher declare that they have no conflicts of interest relevant to the content of this letter.

References

1. Iversen VM, Norum M, Schoenfeld BJ, Fimland MS. No time to lift? Designing time-efficient training programs for strength and hypertrophy: a narrative review. Sports Med. 2021. https://doi.org/10.1007/s40279-021-01490-1.
2. Suchomel TJ, Nimphius S, Stone MH. The importance of muscular strength in athletic performance. Sports Med. 2016;46(10):1419–49.
3. Madrigal L, Robbins JE. Student-athlete stress: an examination in United States collegiate athletics. J Stud Sports Athl Educ. 2020;14(2):123–39.
4. Raedeke TD. Is athlete burnout more than just stress? A sport commitment perspective. J Sport Exerc Psychol. 1997;19(4):396–417.
5. Grandou C, Wallace L, Coutts AJ, Bell L, Impellizzeri FM. Symptoms of overtraining in resistance exercise: international cross-sectional survey. Int J Sports Physiol Perform. 2020;16(1):80–9.
6. Androulakis-Korakakis P, Michalopoulos N, Fisher JP, Keogh J, Loenneke JP, Helms E, Wolf M, Nuckols G, Steele J. The minimum effective training dose required to increase 1RM strength in powerlifters. Front Sport Act Living. 2021;3:248
7. Chilibeck PD, Calder AW, Sale DG, Webber CE. A comparison of strength and muscle mass increases during resistance training in young women. Eur J Appl Physiol Occup Physiol. 1998;77(1–2):170–5.

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LETTER TO THE EDITOR

Response to Comment on: "No Time to Lift? Designing Time-Efficient Training Programs for Strength and Hypertrophy: A Narrative Review"

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Dear Editor,

We thank Carlson et al. [1] for raising some interesting points about our review article on time-efficient resistance training [2] in their letter to the editor. We appreciate the opportunity to respond to their comments and further discussion of this important topic.

The authors of the letter to the editor [1] note that we wrote: "*This narrative review is intended for those in the general public that have limited time for training, and not for those who are seeking to optimize training adaptations without regard to a time commitment (e.g., athletes)*" [2]. Carlson et al. [1] provide several examples of athletic populations with little time for resistance training. We agree with the authors that many athletes (e.g., in various team sports, endurance sports, etc.) may benefit from resistance training, but have limited time for incorporating it into their schedules due to other training, travel, competition, and family obligations. Indeed, many of the principles, methods, and recommendations in our article also apply to athletic populations. It appears that the main issue raised here is that we included the word "athletes" in parentheses. In hindsight we see that this could be misconstrued and acknowledge that although our recommendations were intended for the general public who perceive time as a barrier for exercise, they potentially can apply to athletes as well.

Carlson et al. [1] also note that we wrote *"…it is easier to simulate real-life movements and sport-specific movements with free weights compared to machines, which usually have limited adaptability of the movement pattern"* [2] and thus infer that we "advocate the use of free-weight compared to machine-based resistance exercises" [1]. This is an incorrect inference. In the section titled "Free-Weight and Machine Exercises" [2] we delve into the advantages and disadvantages of both modalities and discuss where they may be appropriate in situation-specific contexts. Moreover, in Table 1 (where we provide a summary of practical applications) we specifically state: "Use machines and/or free weights based on training goals, availability and personal preference" [2].

Finally, Carlson et al. [1] question whether resistance training machines should be recommended over free weights when time efficiency is of primary importance due to the quicker change of loading. We would counter that arguments for time efficiency can be made for both resistance training machines and free weights, and in fact we discuss these nuances in the practical application section of our paper [2].

Declarations

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Conflicts of interest Brad Schoenfeld serves on the scientific advisory board of Tonal Corporation, a manufacturer of fitness equipment. Vegard Iversen, Martin Norum, and Marius Fimland declare that they have no conflicts of interest relevant to the content of this letter.

References

1. Carlson L, Smith D, Fisher JP. Comment on: "No time to lift? Designing time-efficient training programs for strength and

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hypertrophy: a narrative review." Sports Med. 2021. https://doi.org/10.1007/s40279-021-01591-x.
2. Iversen VM, Norum M, Schoenfeld BJ, Fimland MS. No time to lift? Designing time-efficient training programs for strength and hypertrophy: a narrative review. Sports Med. 2021. https://doi.org/10.1007/s40279-021-01490-1.

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